

Part: 05 Documentation

Part 1: Data Exploration

The dataset was explored using basic functions like `head()`, `info()`, and `describe()`. It contains both numerical and categorical features with some missing values. The target variable is `SalePrice`, which is continuous and suitable for regression analysis. The data shows sufficient variation for modeling.

Part 2: Data Cleaning & Preprocessing

Missing values were handled carefully: numerical features were filled with the median to reduce the impact of outliers, while categorical features were filled with “None” to preserve missing information.

Categorical variables were converted into numerical form using one-hot encoding, enabling the model to process them effectively.

Outliers were visualized using boxplots and retained to preserve real-world variability. No feature scaling was applied to maintain model simplicity. The dataset was also checked for duplicates and inconsistencies, with no major issues found.

Part 3: Model Building

The dataset was split into training and testing sets to evaluate model performance on unseen data.

A Linear Regression model was trained to learn relationships between features and house prices. The model was evaluated using RMSE and R^2 score, where RMSE measures prediction error and R^2 indicates how well the model explains variance in the data.

Part 4: Prediction on Test Data

Before making predictions, the training and test datasets were aligned to ensure consistent features after encoding. Missing values in the test set were handled properly. The trained model was then used to generate predictions on unseen data.

Part 5: Submission & Conclusion

Final predictions were saved in a CSV file in the required Kaggle format (Id, SalePrice) and submitted for evaluation.

Overall, the project followed a complete machine learning pipeline including data exploration, cleaning, preprocessing, modeling, and prediction, resulting in a functional regression model for house price estimation.

4667	zhangzelong28		10.91130	3	2d
4668	Alisha Khan #4		11.14512	1	1m
 Your First Entry! Welcome to the leaderboard!					
4669	Underfitted Soul		11.19880	3	2mo